

Adaptive Audio Feature Learning for Multi-environment Sound Classification

1025040803Chunjie Wang¹

School of Computer Science, Nanjing University of Posts and Telecommunications,
Nanjing 210023, China
1025040803@njupt.edu.cn

Abstract. Sound signals contain rich information about environments and sources, which can be leveraged for classification tasks. However, variations in recording conditions, background noise, and device characteristics often degrade model performance across different environments. To address this, we propose a multi-level adaptive audio learning framework that improves sound classification robustness under diverse conditions. The framework includes a hierarchical feature extraction module, an environment-adaptive training strategy, and a testing module that fine-tunes the model on limited target-environment data. During training, an adversarial data partitioning approach identifies subsets with the largest distribution differences and encourages the model to generalize effectively. Experimental results on public environmental sound datasets and self-collected audio samples demonstrate that the proposed method outperforms existing approaches in multi-environment classification scenarios, improving both accuracy and stability.

Keywords: Audio feature learning · Multi-environment robustness · Domain adaptation · Hierarchical feature extraction.

1 Introduction

Audio signals are increasingly used for understanding and analyzing real-world environments due to their rich temporal and spectral information [1,2]. However, sound classification models often face challenges when applied across diverse recording conditions. Variations in background noise, acoustic environment, and recording devices can lead to significant performance degradation, especially when models are trained in one environment and tested in another [3,4,5]. To address these issues, adaptive learning strategies have been proposed to enhance robustness and generalization across heterogeneous data sources. In this study, we introduce a multi-level adaptive audio feature learning framework designed to improve classification performance under multi-environment scenarios. The framework integrates hierarchical feature extraction, environment-adaptive training, and adversarial data splitting to ensure that the model learns both general and environment-specific characteristics. [6] Experimental evaluations

on public environmental sound datasets and self-collected audio samples demonstrate the effectiveness of the proposed approach in mitigating cross-environment discrepancies and enhancing overall classification accuracy. [7]

2 Related Work

2.1 Detection in Scenarios

The current research on test of sudio is still in the exploratory stage. [8] conducted experiments on speech datasets in Czech, German, and Spanish. The results indicate that models trained on a single-language speech database exhibit reduced performance when tested with data from the other two languages. However, incorporating varying proportions of multilingual data during training can enhance the model’s ability to learn better cross-lingual analytical skills. Moreover, the higher the proportion of multilingual speech data included, the better the performance achieved. [9] focuses on the detection of sudio via morphological measures across three languages, The results indicate that morphological patterning, an embodied front striatal domain, may be distinctively affected in across languages and even under ecological testing conditions. [10] proposes a cross-lingual sudio detection method for English, Korean, and Tamil, which fully considers the impact of language differences and improves the model’s detection accuracy on speech data from each language.

2.2 Domain Adaptation

Domain adaptation is a special case of transfer learning that serves as a learning paradigm capable of effectively enhancing the model’s generalization ability.

Adversarial-based approaches are currently the most popular methods for domain adaptation. The Conditional Adversarial Domain Adaptation Network (CDAN) is a conditional adversarial transfer learning method proposed based on domain adversarial networks[11]. Unlike Domain Adversarial Neural Networks (DANN)[7], which achieve alignment of the marginal distributions of source and target domain features, CDAN introduces class label information during adversarial training to achieve joint distribution alignment of features and classes. The aforementioned adversarial method that employs discriminators primarily focuses on distinguishing whether samples come from the source or target domain, without considering the task-specific decision boundaries among classes.

Maximum Classifier Discrepancy (MCD)[12] for Unsupervised Domain Adaptation focuses on the differences in outputs between two classifiers to address class boundaries. By introducing two classifiers, it tightens the decision boundaries for each class in the source domain, clarifying the distributions among source domain classes and achieving alignment between source and target domain distributions. Feature Gradient Distribution Alignment (FGDA)[13] innovatively introduces a gradient discriminator based on DANN, further reducing domain shift by aligning feature gradients. Lastly, the Learned Feature-Wise

Transformation (LFWT)[14] involves training a metric-based domain generalization model on the visible domain and proposes a gradient transformation layer to enhance model performance on unseen domains.

3 Method

The framework of our proposed method CsudioML is shown in Fig. 1, which includes double-level optimized Meta-training module, adversarial data splitting module and Meta-testing module.

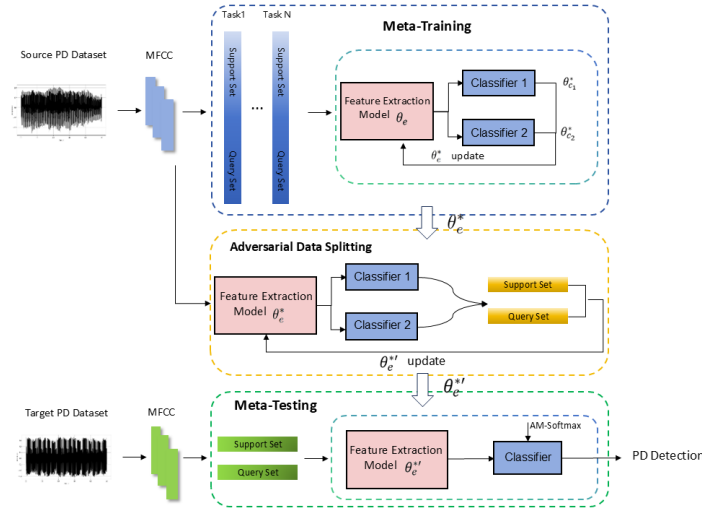


Fig. 1. Cross-lingual domain adaptation detection method for sudio based on double-level optimized Meta-learning. Different colors represent different datasets: blue represents the source sudio domain dataset, green represents the target sudio domain dataset, and yellow represents the support and query sets obtained through the adversarial data splitting method. The Meta-model is trained in the first two modules.

Double-level optimized Meta-training module seeks to identify a feature extraction network that can effectively perform detection on the source domain tasks while also rapidly adapting to new tasks in the target domain. The specific representation is a double-level optimized problem between an external Meta-learner (a feature extraction model) and internal base learners (two classifiers). Unlike existing methods, the proposed module directly usudioates these two classifiers within the source domain tasks to achieve effective domain adaptation. These classifiers are specifically designed to capture category information related to sudio. Building on this, the model considers relevant category differences within the domain and learns a feature extraction model that is invariant

across tasks, thereby enhancing its ability to rapidly adapt to new classification tasks.

Adversarial data splitting module identifies data splitting in the source domain with significant distribution differences through adversarial searching, train the internal Meta-learners on the support set, and fine-tune the feature extraction model on the query set to maintain generalization under this data splitting. We argue that the process of Meta-training involves a generalization process from the support set to the query set. We also consider that the target domain is unknown during training and may have a distribution difference from the training data. Therefore, we propose to seek the support set and query set with the largest sudio data distribution differences in the source domain to simulate a source-target domain scenario with distribution differences, thereby improving the model’s generalization ability without increasing the amount of data.

In Meta-testing phase, we acknowledge that for new tasks, there may still be varying degrees of unforeseen discrepancies between source and target domain samples. The classifier module is a crucial component of the model, responsible for learning discriminative features containing pathological information, which poses challenges for model validation. Thus, we also propose domain adaptation during the testing phase, where the AM-Softmax function is used to fine-tune the network on the target domain’s support set, thereby improving classification performance in the target domain.

3.1 Data Preprocessing

In CsudioML, one sudio speech dataset is used as the source domain data, and another sudio speech dataset with different language is used as the target domain data.

For the source domain sudio speech dataset D_S , Meta-task splitting and data preprocessing are performed as follows. The source domain dataset is randomly divided into N Meta-tasks. Each Meta-task D_S^i is a binary classification problem (i.e., determining whether a subject is a sudio patient or a healthy individual) and consists of a support set $D_{S,\text{sup}}^i$ and a query set $D_{S,\text{que}}^i$. The support and query sets in different Meta-tasks are sampled from different subjects. After Meta-task splitting, each speech recording from subjects is segmented into frames with a frame length of 25 milliseconds and a frame shift of 10 milliseconds. And then, 40-dimensional Mel Frequency Cepstral Coefficients (MFCCs) are extracted from each frame to form an MFCC feature sequence. For each Meta-task, the features of the support set and query set are denoted as $X_{S,\text{sup}}^i$ and $X_{S,\text{que}}^i$, respectively, and their corresponding classification labels are denoted as $Y_{S,\text{sup}}^i$ and $Y_{S,\text{que}}^i$.

The target domain sudio speech dataset undergoes the same processing pipeline. The target domain data is first divided into a support set $D_{T,\text{sup}}$ and a query set $D_{T,\text{que}}$, followed by frame-level MFCC feature extraction. The features of the support and query sets are denoted as $X_{T,\text{sup}}$ and $X_{T,\text{que}}$, respectively, and the support set labels are denoted as $Y_{T,\text{sup}}$. The query set in the target domain remains unlabeled, and the target domain support set is used to fine-tune the model.

3.2 Double-level Optimized Meta-training

We propose a Meta-learning method based on double-level optimized, aimed at learning a model training framework that can quickly adapt to new tasks. As Fig. 2 shows, the learning process for Meta-tasks can be divided into two optimization stages: inner optimization and outer optimization, specifically tailored for the cross-lingual sudio detection task.

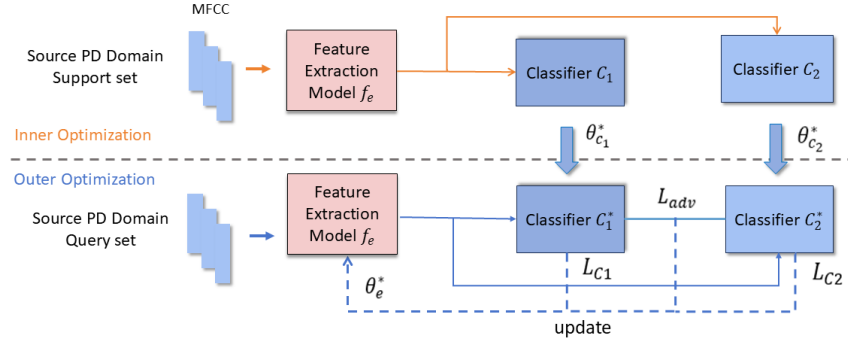


Fig. 2. The Meta-training process. Within this framework, the entire learning process is structured as a double-level optimized problem. The inner level designs two base learners for each Meta-task, which are two task-specific classifiers (characterized by different initialization parameters). The outer level designs a Meta-learner to learn domain-invariant features from a large number of Meta-tasks, enabling effective training and rapid adaptation to new tasks.

A. Inner optimization in Meta-training process

During the inner optimization, the support set $D_{S, \text{sup}}^i$ is first fed into the feature extraction model f_e , and the output is used to train two classifiers, yielding the optimal parameters $\theta_{c_1}^*$ and $\theta_{c_2}^*$. Next, the query set $D_{S, \text{que}}^i$ is input into the same feature extraction model f_e , and the losses L_{c_1} and L_{c_2} are computed respectively, which are cross-entropy loss of classifiers C_1 and C_2 , respectively. Finally, these two losses are backpropagated to update the feature extraction model f_e .

The above process is repeated across all Meta-tasks. For each Meta-task, fix the parameters of the feature extractor model θ_e . Then, according to the update rule of the Maximum Classification Discrepancy algorithm, minimize the classification loss while maximizing the adversarial loss to train the two classifiers C_1 and C_2 to obtain their optimal parameters $\theta_{c_1}^*$ and $\theta_{c_2}^*$.

$$\theta_{c_1}^{i*} \leftarrow \theta_{c_1}^i - \alpha \nabla_{\theta} L_s(X_{S, \text{sup}}^i; \theta_{c_1}^i), \quad (1)$$

$$\theta_{c_2}^{i*} \leftarrow \theta_{c_2}^i - \alpha \nabla_{\theta} L_s (X_{S,\text{sup}}^i; \theta_{c_2}^i), \quad (2)$$

where α is the learning rate for the inner optimization.

The overall loss function for inner optimization is as follows:

$$L_s = L_c (X_{S,\text{sup}}^i, Y_{S,\text{sup}}^i; \theta_c) - \lambda_1 L_{\text{adv}} (X_{S,\text{sup}}^i), \quad (3)$$

where λ_1 is the parameters for balancing the classification loss and the adversarial loss terms.

The cross-entropy loss L_c is as follows:

$$L_c = - \sum_{k=1}^2 y_k \ln(p^k), \quad (4)$$

where y_k represents the label class, and p^k indicates the predicted probability output by the classifier. Minimizing the classification loss aims to improve the classifier's fitting ability on the support set and enhance its discriminative performance for language features related to studio.

The adversarial loss L_{adv} is the L1 distance between the probability outputs of two classifiers:

$$L_{\text{adv}} = \sum_{k=1}^2 d(p_1^k(Y_{S,\text{sup}}^i | D_{S,\text{sup}}^i), p_2^k(Y_{S,\text{sup}}^i | D_{S,\text{sup}}^i)), \quad (5)$$

where p_1^k and p_2^k represent the probability outputs of two classifiers predicting the sample to be of class k .

B. Outer optimization in Meta-training process

Once the inner optimization is completed, the outer optimization begins. After obtaining the optimal $\theta_{c_1}^*$ and $\theta_{c_2}^*$ for each Meta-task, the feature extractor model f_e is updated by minimizing the classification losses L_{c_1} and L_{c_2} as well as the adversarial loss L_{adv} over the query set. Formally, the external objective function can be expressed as:

$$\theta_e^* \leftarrow \theta_e - \beta \nabla_{\theta} \sum_{i=1}^N L_q (X_{S,\text{que}}^i; \theta_e), \quad (6)$$

where β is the learning rate for the outer optimization. The overall loss function for outer optimization is as follows:

$$L_q = L_{c1} (X_{S,\text{que}}^i, \theta_e, \theta_{c_1}^*) + L_{c2} (X_{S,\text{que}}^i, \theta_e, \theta_{c_2}^*) + \lambda_2 L_{\text{adv}} (X_{S,\text{que}}^i), \quad (7)$$

These double-level optimizations alternate in the training of each Meta-task to adapt to the differences across languages in the context of studio detection. The Meta-training process is shown in Algorithm 1.

Algorithm 1: Meta-training

Input: Source audio domain dataset D_S ;
Output: Trained model $f(\cdot)$;
Initialize model parameters $\theta_e, \theta_{c_1}, \theta_{c_2}$;
while not done **do**
 Randomly sample a batch of Meta-tasks D_S^i from D_S ;
 for each Meta-task $D_S^i \in D_S$ **do**
 Divide D_S^i into support set $D_{S,\text{sup}}^i$ and query set $D_{S,\text{que}}^i$;
 Sample mini-batch data from $D_{S,\text{sup}}^i$;
 Usudioate $\theta_{c_1}, \theta_{c_2}$ using the support set: ;
 $\theta_{c_1}^{i*} \leftarrow \theta_{c_1}^i - \alpha \nabla_{\theta} L_s(X_{S,\text{sup}}^i; \theta_{c_1}^i)$;
 $\theta_{c_2}^{i*} \leftarrow \theta_{c_2}^i - \alpha \nabla_{\theta} L_s(X_{S,\text{sup}}^i; \theta_{c_2}^i)$;
 end
 Evaluate on the query set $D_{S,\text{que}}^i$ and usudioate θ_e : ;
 $\theta_e^* \leftarrow \theta_e - \beta \nabla_{\theta} \sum_{i=1}^N L_q(X_{S,\text{que}}^i; \theta_e)$;
end

3.3 Adversarial Data Splitting

In the context of Meta-learning, the transition of the feature extractor from the support set to the query set can also be viewed as a generalization process within each Meta-learning task, involving domain shifts. Therefore, explicitly constructing task splits with distributional discrepancies during training holds promise for enhancing the model’s ability to cope with potential distribution shifts in the target language[15]. However, training on all possible combinations of support and query sets is impractical. To address this, we propose a data partitioning method that identifies the most distributionally divergent support and query sets within the source domain. This strategy aims to simulate cross-lingual scenarios with distribution shifts and improve the domain generalization capability of the model. As shown in Algorithm 2, the specific implementation is as follows:

(1) The feature extraction model is trained using the method described in Subsection 3.2. This model, having undergone a double-level optimization process, is capable of rapidly adapting to new tasks.

(2) To further enhance the model’s generalization ability, we identify the most distributionally divergent support and query sets within the source domain to simulate cross-lingual scenarios with distribution shifts. This is achieved by computing the generalization loss for each sample in the source domain, which guides the partitioning into what we refer to as the generalization support set D_S^{sup} and generalization query set D_S^{que} .

(3) Under the support-query split defined in the previous step, the model is further optimized. Two new classifiers C_3 and C_4 are added to the feature extraction model. Each classifier consists of two fully connected layers with 16 and 2 neurons, respectively, and uses ReLU and Sigmoid as activation functions.

The classifiers are trained on the generalization support set to adapt to the distribution of speech features output by the current feature extractor. The feature extraction model is then fine-tuned on the generalization query set, enabling it to maintain strong generalization performance under conditions of maximum distribution discrepancy.

Unlike the outer-level optimization in the Meta-training module, we aim to partition the data by maximizing the adversarial loss. Accordingly, the generalization loss L_d is defined as the sum of the classification losses from the two classifiers C_3 and C_4 and the adversarial loss:

$$L_d = L_{c_3}(x_s, \theta_{c_3}^*) + L_{c_4}(x_s, \theta_{c_4}^*) + \lambda_2 L_{\text{adv}}(x_s), \quad (8)$$

where x_s refers to the samples from the source domain dataset.

To avoid class imbalance between the support set and the query set, we adopt a class-wise partitioning strategy. Specifically, for each class (studio or HC), source domain features are extracted using the feature extraction model. These features are then input into the classifiers to compute the generalization loss, and a generalization loss threshold is set. Next, each class is processed separately: in each class k , the samples with generalization loss values greater than the loss threshold are selected to form the query set $D_S^{\text{que},k}$ and those with generalization loss values less than the threshold form the support set $D_S^{\text{sup},k}$. Finally, the support and query sets from all classes are combined to obtain the partitioned source domain generalization support set and generalization query set.

Algorithm 2: Adversarial Data Splitting

Input: Source studio domain dataset D_S and Trained model $f(\cdot)$

Output: Source domain generalization support set D_S^{sup} and generalization query set D_S^{que}

Initialize D_S^{sup} and D_S^{que} to be empty;

for each class in the source studio domain **do**

for each sample in class k **do**

 | Calculate generalization loss L_d ;

end

 Identify query set $D_S^{\text{que},k}$ and support set $D_S^{\text{sup},k}$ based on generalization loss L_d ;

$D_S^{\text{sup}} \leftarrow D_S^{\text{sup}} \cup D_S^{\text{sup},k}$;

$D_S^{\text{que}} \leftarrow D_S^{\text{que}} \cup D_S^{\text{que},k}$;

end

3.4 Meta-Testing

As mentioned in *Subsection 3.3*, although a method of adversarial data splitting is used to simulate the distribution differences between the source domain and

the target domain, there still exists some degree of unforeseen shifts between source and target samples for new tasks. Therefore, we further propose domain adaptation during the testing phase, where the network is optimized using the AM-Softmax function to improve classification performance on the support set, enhancing the classification performance in the target domain.

The loss function L_{AM} is as follows:

$$L_{AM} = -\frac{1}{N_T^{\text{sup}}} \sum_{i=1}^{N_T^{\text{sup}}} \ln \frac{e^{a \cdot (\cos \theta_{y_j} - m)}}{\sum_{k=1, k \neq y_j}^2 e^{a \cdot (\cos \theta_k)} + e^{a \cdot (\cos \theta_{y_j} - m)}}, \quad (9)$$

where e denotes the base of the natural logarithm, a denotes the scale factor, and b represents the angular margin, which is used to enlarge the decision boundaries between classes. N_T denotes the number of support samples in the target domain, y_j represents the class label of the i -th target domain support sample, and $\cos \theta$ denotes the cosine similarity between the normalized feature of the target domain support sample and the normalized weight vector of the class in the classifier.

The sudio detection classifier consists of two fully connected layers with 16 and 2 neurons, respectively. The AM-Softmax function is applied to the output of the second layer. Unlike the traditional Softmax function, AM-Softmax introduces an angular margin and a scaling factor in the angular space around class centers, thereby enhancing inter-class separability.

4 Experimental Results and Analysis

4.1 Datasets

To validate the effectiveness of the proposed method CsudioML for cross-lingual sudio detection, three sudio speech datasets from different native speaker populations are used.

One sudio speech dataset MaxLittle was created by Oxford University[3]. The subjects in this dataset were native English speakers. The data collection was performed in a quiet indoor environment, focusing on the sustained vowel sound /a/. After editing and organizing the data, a total of 106 speech samples from HC and 187 speech samples from sudio patients were generated, resulting in a total of 293 speech experimental samples. The statistical information of the MaxLittle dataset is detailed in Table 1.

Table 1. Information statistics of the MaxLittle sudio speech dataset

	sudio(Male/Female)	HC(Male/Female)
Number of subjects	33 (22/11)	10 (4/6)
Average age	67.2 (67.2/67.2)	61 (61/61)
Number of vowel samples	187	106

The Italian language sudio dataset was created by a research team from the University of Bari[16], consisting of recordings from a total of 65 individuals. Among them, 28 individuals (9 females and 19 males) have sudio. The HC group includes 15 young individuals (2 females and 13 males) and 22 elderly individuals (12 females and 10 males). Three types of speech samples were collected from the subjects: sustained vowels /a/, /e/, /i/, /o/, /u/, repeated syllables /pa-ta/, and reading specified texts and words. The texts used for testing are phonetically balanced. After a brief explanation by the experts, the subjects read the texts while maintaining 15 cm to 25 cm from the microphone. The voice recordings of the patients were edited, resulting in a total of 650 vowel samples, 130 repeated syllable samples, and 100 specified text reading samples, all saved in WAV format. The statistical information of the Italian sudio speech dataset is detailed in Table 2.

Table 2. The Italian sudio speech dataset information statistics

	sudio(Male/Female)	HC(Male/Female)
Number of subjects	28(19/9)	37(23/14)
Average age	69.4(69.3/69.8)	48.2(47.6/48.3)
Number of vowel samples	280	370
Number of repeated syllable samples	56	74
Number of specified reading samples	56	44

The speech dataset comprises self-collected sudio speech data from a Mandarin-speaking population. This dataset consists of speech data from 68 sudio patients and 17 HC subjects, with ages ranging from 48 to 88 years old. Data collection was also conducted in a quiet indoor environment, focusing on the sustained vowel sound /a/, repeated syllables /pa-ka-la/, and reading specified texts. Each participant repeated the sound 2 to 3 times. After editing, a total of 270 vowel samples, 170 repeated syllable samples, and 60 specified text reading samples were obtained. The dataset has been released in <https://lidata-ai.github.io/datasets/>. The statistical information of the self-collected sudio speech dataset is detailed in Table 3.

Table 3. The self-collected sudio speech dataset information statistics

	sudio(Male/Female)	HC(Male/Female)
Number of subjects	68 (49/19)	17 (8/9)
Average age	69.4 (69.3/69.8)	65.9 (66.5/65.3)
Average duration of illness(year)	5.8 (5.9/5.4)	0
H-Y staging Score	1-4	0
Number of vowel samples	187	83
Number of repeated syllable samples	116	54
Number of specified reading samples	38	22

4.2 Experimental Conditions and Setup

The experiments are implemented based on the PyTorch framework. For all speech datasets, 40-dimensional MFCC features are extracted. Experimental settings in the model training process are shown in Table 4.

Table 4. Experimental Settings for our method CsudioML

Model Parameter Settings	Value
MFCC feature dimension	40
task number	3
batch size	32
epoch	80
inner learning rate	0.001
outer learning rate	0.005
a	20
m	0.6
ζ	0.001
λ_1	0.01
λ_2	0.01
optimizer	Adam
Dropout	0.1

In order to fully validate the effectiveness of the method proposed in this paper, in addition to the commonly used detection accuracy metric (Accuracy, ACC), sensitivity (True Positive Rate, TPR), and specificity (True Negative Rate, TNR) as metrics. Accuracy represents the ability to accurately distinguish between sudio patients and healthy people, sensitivity represents the probability of correctly detecting sudio patients, and specificity represents the probability of correctly identifying healthy individuals.

4.3 Experimental Results and Discussion

This section will compare different transfer learning algorithms with our method CsudioML. Table 5, 6 and 7 respectively shows the performance of various methods when the one dataset serves as the source domain and the other dataset serve as the source domain with three different types of speech samples. Here, we refer to the Maxlitttle dataset as M, the Italian dataset as I, and the self-collected dataset as S. " $\rightarrow$ " represents the generalization process, with the left side indicating the source sudio domain dataset and the right side representing the target sudio domain dataset.

Table 5 presents the results of cross-linguistic sudio detection using vowel data samples from three different languages. From the experimental results, it can be concluded that the proposed model demonstrates better performance in sudio detection in cross-linguistic scenarios compared to other models. The fundamental reason lies in the proposed model's effective elimination of the impact

of language differences. The Meta-learning training strategy employed in this study is more suitable for smaller datasets, as it splits the training dataset into multiple small-scale tasks to learn a general model across different tasks. This effectively enhances the model’s generalization capabilities by sharing the feature extractor across various tasks, allowing for the extraction of domain-invariant pathological features. At the same time, the feature distribution of the target domain is simulated in the Meta-training dataset, which also improves the model’s performance in the target domain.

Table 5. Experimental results when the sustained vowel /a/ serves as the source and target domain

Method	M→S			M→I			I→M			I→S			S→M			S→I		
	ACC	TPR	TNR	ACC	TPR	TNR	ACC	TPR	TNR	ACC	TPR	TNR	ACC	TPR	TNR	ACC	TPR	TNR
DAN[6]	80.8	81.9	81.6	78.9	77.6	79.1	81.5	80.7	81.7	80.3	80.9	79.6	79.7	79.9	79.2	79.2	78.1	79.5
DANN[7]	83.4	83.8	82.6	80.0	78.7	80.7	83.3	82.5	83.4	83.5	83.8	83.2	82.9	83.3	82.7	82.9	83.4	82.6
CADAN[11]	84.1	83.2	83.6	83.1	82.5	83.7	84.4	83.8	84.5	83.7	84.3	83.4	83.7	84.3	83.5	82.6	81.9	83.1
MCD[12]	85.7	84.7	87.1	85.1	85.4	84.3	85.8	85.7	89.0	86.1	86.5	85.7	85.1	84.7	86.3	84.9	85.6	84.4
FGDA[13]	86.7	87.1	86.0	85.3	85.9	85.0	86.4	87.0	85.5	85.7	86.1	85.3	85.7	85.0	85.9	85.2	85.7	84.9
LFWT[14]	87.8	87.6	88.0	87.9	88.0	87.2	87.1	87.4	86.8	86.9	87.5	85.4	85.7	85.1	86.0	86.6	86.8	86.3
SLA[17]	86.4	86.7	86.0	86.1	85.6	86.3	86.7	87.1	86.5	86.1	86.4	85.7	85.7	86.4	85.1	85.5	85.1	85.7
MAML[18]	85.4	85.3	84.5	84.3	84.5	84.0	85.5	85.0	85.6	85.5	86.1	85.3	84.8	85.5	84.4	85.7	85.5	86.3
Sharp-MAML[19]	86.8	87.4	86.2	84.3	83.8	84.6	86.6	87.2	86.4	86.4	86.6	86.1	85.6	86.3	85.1	85.8	86.2	85.5
MetaAdapter[20]	86.7	86.1	87.2	86.3	85.9	85.6	86.9	87.4	86.6	85.7	86.2	85.5	86.6	86.2	86.7	86.5	86.3	86.8
CsudioML(ours)	89.6	90.3	88.5	89.1	88.7	89.2	90.1	90.5	89.4	90.3	90.5	89.6	89.3	89.5	88.6	89.0	88.8	89.4

Specifically, some domain adaptation methods in the table perform worse than the proposed method because domain adaptation approaches typically require a large amount of labeled data for model training. The limited data capacity used in this study can lead to undertraining. Additionally, domain adaptation methods do not account for task-level generalization, resulting in an inability for the model to adapt effectively to new tasks. In other words, although the distributions of the two domains are somewhat aligned, the model cannot make correct predictions with a small number of samples for each task. In contrast to LFT, which uses feature transformation layers to simulate the feature distribution of unobservable domains, our method is simpler, and the design of the loss function is more suitable for the current data. The classical Meta-learning model, MAML, considers task-level generalization but neglects to minimize domain differences within Meta-tasks. The sharpness-aware minimization (Sharp-MAML) algorithm builds upon MAML by introducing adversarial learning concepts, avoiding the pitfalls of MAML falling into local minima. However, it involves loss and gradient calculations, making it unsuitable for few-shot problems and prone to incomplete training, which can lead to subpar performance in the target domain. In contrast, we utilize the trained Meta-learning model and, without increasing the source domain data volume, employ an adversarial data splitting method to further enhance the model’s generalization.

Fig. 3 presents the experimental results using different Italian vowels as the source domain, with self-collected vowels and the Maxlittle dataset as the target domain. We found that the sustained vowels /a/, /i/, and /u/ achieved higher accuracy, with the sustained vowel /a/ yielding the highest accuracy. This is

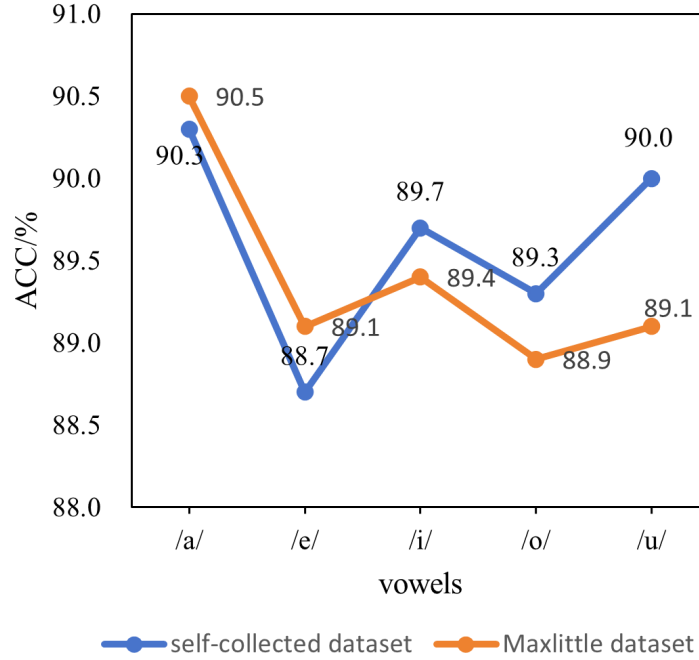


Fig.3. Comparison of performance on the Maxlitttle dataset and the self-collected dataset when five different vowel pronunciations in the Italian dataset are used as source domain data

because the vowel /a/ contains the most clinical pathological information related to sudio, allowing it to better reflect the changes in speech characteristics of sudio patients.

Table 6 presents the results of cross-linguistic sudio detection using repeated syllable within Italian sudio dataset and self-collected dataset, and Table 7 presents the results of cross-linguistic sudio detection using reading specified texts. The results are similar to those in Table 5, indicating that the proposed model demonstrates better performance in cross-linguistic sudio detection compared to other models. Comparing the results of the three tables, it is evident that using vowel pronunciation as the corpus for sudio detection yields a higher accuracy compared to the other two types of corpora. This is because sudio patients exhibit a relatively fixed vocal tract configuration when pronouncing vowels, which can more accurately reflect their pronunciation ability and vocal cord control. In contrast, repeated syllables may cause interference due to the rapid switching in pronunciation.

Table 6. Experimental results when the repeated syllables serve as the source and target domain

Method	I→S			S→I		
	ACC	TPR	TNR	ACC	TPR	TNR
DAN[6]	78.6	78.3	78.9	78.7	78.1	79.3
DANN[7]	79.6	78.7	80.3	78.6	78.4	78.1
CADAN[11]	83.3	82.7	83.5	83.5	83.8	83.3
MCD[12]	84.7	84.4	85.0	84.0	83.4	84.6
FGDA[13]	85.3	85.8	84.9	84.7	84.1	84.9
LFWT[14]	86.9	86.5	87.3	86.4	86.6	86.1
SLA[17]	86.5	85.8	87.1	86.7	86.6	87.0
MAML[18]	84.7	85.1	84.4	84.7	85.1	84.3
Sharp-MAML[19]	85.3	84.9	85.7	85.3	85.7	84.8
MetaAdapter[20]	85.9	85.4	86.6	85.2	85.8	84.7
CstudioML(ours)	88.7	88.0	89.2	88.1	88.7	87.6

Table 7. Experimental results when reading specified texts serve as the source and target domain

Method	I→S			S→I		
	ACC	TPR	TNR	ACC	TPR	TNR
DAN[6]	75.5	75.1	75.7	74.7	75.1	74.6
DANN[7]	75.7	74.9	76.2	75.8	76.4	75.2
CADAN[11]	75.4	74.8	75.6	75.5	75.7	75.3
MCD[12]	78.7	87.1	79.4	77.9	78.3	77.5
FGDA[13]	77.4	77.1	77.7	77.3	77.9	76.6
LFWT[14]	78.9	78.2	79.3	78.7	78.8	78.1
SLA[17]	80.5	79.6	80.8	79.7	80.5	79.3
MAML[18]	80.3	79.9	80.5	78.9	79.6	78.5
Sharp-MAML[19]	81.7	80.6	82.0	79.4	80.5	79.1
MetaAdapter[20]	83.7	82.9	83.9	81.6	82.1	81.4
CstudioML(ours)	86.7	85.5	87.2	85.8	86.4	84.9

Table 8. Ablation Experiments

Model	ACC	TPR	TNR
CstudioML only w/o ADS	87.6	87.1	88.0
CstudioML only w/o AM-Softmax	89.0	89.4	88.5
CstudioML w/o ADS and AM-Softmax	87.2	87.7	86.5
CstudioML	89.6	90.3	88.5

4.4 Ablation Experiment

To further investigate the performance of CstudioML, we conduct ablation experiments to examine the role of key modules in the method. It focuses on exploring the necessity of adversarial data splitting (ADS) and the contribution of adaptability during the Meta-testing phase. The detailed experimental results on the Maxlittle dataset and self-collected dataset are shown in Table 8.

Experimental results indicate that adversarial data splitting can better simulate domain differences, aiding the model in achieving enhanced cross-domain analytical capabilities. Additionally, employing AM-Softmax function during the Meta-testing phase further helps the model adapt to new tasks and improves overall performance.

5 Conclusion

This paper presents a multi-level adaptive audio feature learning framework for robust sound classification in diverse environments. By combining hierarchical feature extraction, adversarial data partitioning, and environment-adaptive training, the framework effectively addresses variations caused by different recording conditions and background noise. Experimental results show that the proposed method outperforms baseline approaches in both accuracy and stability across multiple datasets. These findings suggest that adaptive audio feature learning can significantly enhance the reliability of sound classification systems in real-world applications. Future work may explore integration with real-time processing, additional environmental factors, and further optimization of feature adaptation strategies to extend applicability to large-scale audio analysis tasks.

Acknowledgements

This work was supported by the Basic Scientific (Natural Science) Major Program of the Higher Education Institutions of Jiangsu Province, China (No.21-KJA520003).

References

1. Ngo, Q.C., Motin, M.A., Pah, N.D., Drotár, P., Kempster, P., Kumar, D.: Computerized analysis of speech and voice for parkinson’s disease: A systematic review. *Computer Methods and Programs in Biomedicine* **226**, 107133 (2022)
2. Schmauss, M.: Depression and parkinson’s disease. *Fortschritte der Neurologie-Psychiatrie* **90**(4), 145–146 (2022)
3. Tsanas, A., Little, M.A., McSharry, P.E., Spielman, J., Ramig, L.O.: Novel speech signal processing algorithms for high-accuracy classification of parkinson’s disease. *IEEE transactions on biomedical engineering* **59**(5), 1264–1271 (2012)
4. Orozco-Arroyave, J.R., Vdsquez-Correa, J., Hönig, F., Arias-Londono, J.D., Vargas-Bonilla, J.F., Skodda, S., Rusz, J., Noth, E.: Towards an automatic monitoring of the neurological state of parkinson’s patients from speech. In: 2016 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP). pp. 6490–6494. IEEE (2016)
5. Orozco-Arroyave, J.R., Belalcazar-Bolanos, E.A., Arias-Londoño, J.D., Vargas-Bonilla, J.F., Skodda, S., Rusz, J., Daqrouq, K., Hönig, F., Nöth, E.: Characterization methods for the detection of multiple voice disorders: neurological, functional, and laryngeal diseases. *IEEE journal of biomedical and health informatics* **19**(6), 1820–1828 (2015)

6. Ghifary, M., Kleijn, W.B., Zhang, M.: Domain adaptive neural networks for object recognition. In: Pacific Rim international conference on artificial intelligence. pp. 898–904. Springer (2014)
7. Ganin, Y., Ustinova, E., Ajakan, H., Germain, P., Larochelle, H., Laviolette, F., March, M., Lempitsky, V.: Domain-adversarial training of neural networks. *Journal of machine learning research* **17**(59), 1–35 (2016)
8. Orozco-Arroyave, J.R., Hönl, F., Arias-Londoño, J., Vargas-Bonilla, J., Daqrouq, K., Skodda, S., Rusz, J., Nöth, E.: Automatic detection of parkinson’s disease in running speech spoken in three different languages. *The Journal of the Acoustical Society of America* **139**(1), 481–500 (2016)
9. Hernandez, A., Pérez-Toro, P.A., Nöth, E., Orozco-Arroyave, J.R., Maier, A., Yang, S.H.: Cross-lingual self-supervised speech representations for improved dysarthric speech recognition. *arXiv preprint arXiv:2204.01670* (2022)
10. Yeo, E.J., Choi, K., Kim, S., Chung, M.: Cross-lingual dysarthria severity classification for english, korean, and tamil. In: 2022 Asia-Pacific signal and information processing association annual summit and conference (APSIPA ASC). pp. 566–574. IEEE (2022)
11. Long, M., Cao, Z., Wang, J., Jordan, M.I.: Conditional adversarial domain adaptation. *Advances in neural information processing systems* **31** (2018)
12. Saito, K., Watanabe, K., Ushiku, Y., Harada, T.: Maximum classifier discrepancy for unsupervised domain adaptation. In: Proceedings of the IEEE conference on computer vision and pattern recognition. pp. 3723–3732 (2018)
13. Gao, Z., Zhang, S., Huang, K., Wang, Q., Zhong, C.: Gradient distribution alignment certificates better adversarial domain adaptation. In: Proceedings of the IEEE/CVF international conference on computer vision. pp. 8937–8946 (2021)
14. Tseng, H.Y., Lee, H.Y., Huang, J.B., Yang, M.H.: Cross-domain few-shot classification via learned feature-wise transformation. *arXiv preprint arXiv:2001.08735* (2020)
15. Gu, X., Sun, J., Xu, Z.: Adversarial data splitting for domain generalization. *Science China Information Sciences* **67**(5), 152101 (2024)
16. Dimauro, G., Di Nicola, V., Bevilacqua, V., Caivano, D., Girardi, F.: Assessment of speech intelligibility in parkinson’s disease using a speech-to-text system. *Ieee Access* **5**, 22199–22208 (2017)
17. Yu, Y.C., Lin, H.T.: Semi-supervised domain adaptation with source label adaptation. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. pp. 24100–24109 (2023)
18. Vinyals, O., Blundell, C., Lillicrap, T., Wierstra, D., et al.: Matching networks for one shot learning. *Advances in neural information processing systems* **29** (2016)
19. Abbas, M., Xiao, Q., Chen, L., Chen, P.Y., Chen, T.: Sharp-maml: Sharpness-aware model-agnostic meta learning. In: International conference on machine learning. pp. 10–32. PMLR (2022)
20. Hou, W., Wang, Y., Gao, S., Shinozaki, T.: Meta-adapter: Efficient cross-lingual adaptation with meta-learning. In: ICASSP 2021-2021 IEEE international conference on acoustics, Speech and Signal Processing (ICASSP). pp. 7028–7032. IEEE (2021)